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Starlinked! An analysis of SpaceX's small satellite mega-constellation under the Fourth Amendment

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ABSTRACT
Private commercialization is making broadband internet accessible through a new frontier: space. Space Exploration Holdings, LLC (SpaceX), is launching a mega-constellation of thousands of small-satellites into low-Earth orbit to provide fixed-satellite service across the globe. In order to do so, SpaceX was required to receive a license from the Federal Communications Commission (FCC). The licensing regime is part of a larger international regulatory framework based on five governing treaties – the Outer Space Treaty, Rescue Agreement, Liability Convention, Registration Convention, and the Moon Treaty. Compliance with these treaties is managed through the International Telecommunications Union (ITU) an agency of the United Nations. In this instance, the FCC's domestic licensing processes ensure compliance by the United States to the ITU's procedures and the treaties. The role of the FCC in maintaining these regulatory operations may, however, introduce risks to consumer privacy. This paper explores the tension between government regulation of information and communication technologies, specifically, small satellite mega-constellations and the right to privacy under the Fourth Amendment of the United States Constitution.

KEYWORDS
Satellites; Starlink; SpaceX; Federal Communications Commission; Fourth Amendment; Outer Space Treaty; communications law

Introduction

While space travel once seemed like an unrealistic idea, private commercialization has quickly made it a reality. Pioneering efforts by Elon Musk's Space Exploration Holdings, LLC (SpaceX)[1] has catalyzed a new generation of space exploration missions. One such initiative by SpaceX is the Starlink program. Starlink's small satellite mega-constellation will enable internet accessibility in remote locations.[2] While Starlink is an innovative initiative, it is still subject to international and domestic law,[3] and the governing regulatory framework is not insulated from legal challenges – including questions of liability, and national security.

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[1]33 FCC Rcd 3391 (4) para. 1.

[2]Adam Mann, 'Starlink: SpaceX's Satellite Internet Project' (Space.com, Jan. 17, 2020) <https://www.space.com/spacex-starlink-satellites.html>.

[3]47 USC § 301 (d), (f); 47 USC § 303(r).

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This paper will explore the international and domestic regulatory framework governing small satellites in low-Earth orbit (LEO), and discuss whether the United States government could legally access consumer data collected by private commercial satellites such as Starlink.

I. Background

This section provides an overview of relevant international and domestic regulatory authorities and processes, as well as in-depth review of the approval process for Starlink and its impact on small-satellite licensing. Then, this section shall provide an overview of Fourth Amendment jurisprudence as it relates to aerial imaging, the third-party doctrine, and national security.

A. International regulatory background – five governing treaties

1. Outer Space Treaty

In 1967, the Treaty on the Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies ('Outer Space Treaty') was entered into force.[4] Article VI of the treaty provides that states to the treaty will be responsible for any activity by government agencies or non-government agencies.[5] Further, activities of non-governmental entities in outer space require authorization and supervision by the appropriate State Party to the treaty.[6] Finally, when an international organization carries out activities in outer space, both the international organization and the state parties to the treaty are responsible for compliance with the Outer Space Treaty.[7]

Article VII vests liability of any damage occurring on Earth, in air space, and outer space on the launch authorizing State Party.[8] Each State Party to the Treaty that launches or procures the launching of an object into outer space, and each State Party from whose territory or facility an object is launched, is internationally liable for damage to another State Party to the Treaty or to its natural or juridical persons by such object or its component parts on the Earth, in air space or in outer space, including the moon and other celestial bodies.[9]

[4]Treaty on the Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies (opened for signature 27 January 1967, entered into force 10 October 1967) 610 UNTS 205, 18 UST 2410 (Outer Space Treaty).

[5]ibid art. 6 ('States Parties to the Treaty shall bear international responsibility for national activities in outer space, including the moon and other celestial bodies, whether such activities are carried on by governmental agencies or by non-governmental entities, and for assuring that national activities are carried out in conformity with the provisions set forth in the present Treaty. The activities of non-governmental entities in outer space, including the moon and other celestial bodies, shall require authorization and continuing supervision by the appropriate State Party to the Treaty. When activities are carried on in outer space, including the moon and other celestial bodies, by an international organization, responsibility for compliance with this Treaty shall be borne both by the international organization and by the States Parties to the Treaty participating in such organization.').

[6]ibid.

[7]ibid.

[8]ibid art. 7 ('Each State Party to the Treaty that launches or procures the launching of an object into outer space, including the moon and other celestial bodies, and each State Party from whose territory or facility an object is launched, is internationally liable for damage to another State Party to the Treaty or to its natural or juridical persons by such object or its component parts on the Earth, in air space or in outer space, including the moon and other celestial bodies.').

[9]ibid.

2. Rescue Agreement

The Agreement on The Rescue of Astronauts, Return of Astronauts and Return of Objects Launched into Outer Space is known as the 'Rescue Agreement'.[10]

3. Liability Convention

The Convention on International Liability for Damage Caused by Space Objects ('Liability Convention') establishes liability for 'launching states'.[11] The Liability Convention provides that '[a] launching State shall be absolutely liable to pay compensation for damage caused by its space object on the surface of the earth or to aircraft flight'.[12] Article III of the Liability Convention establishes liability for damage caused elsewhere than on the surface of earth, and clarifies that liability shall only be extended to a launching state if the damage is 'due to its fault or the fault of persons for whom it is responsible'.[13]

Article IV details provisions for liability where two states would be jointly and severally liable to a third state.[14] Article VI provides terms for exoneration from liability for the launching state, and does not extend exoneration to a State that is not in conformity with international law, such as the Outer Space Treaty.[15] Article VII similarly provides exemptions to damage caused by a space object of a launching state to nationals of

[10]Agreement on the Rescue of Astronauts, the Return of Astronauts and the Return of Objects Launched into Outer Space (adopted 19 December 1967, entered into force 3 December 1968) 19 UST 7570 (Rescue Agreement) United Nations Office for Outer Space Affairs <https://www.unoosa.org/oosa/en/ourwork/spacelaw/treaties/introrescueagreement.html> accessed 10 January 2020.

[11]Convention on International Liability for Damage Caused by Space Objects (opened for signature 29 March 1972, entered into force signed September 1972) 961 UNTS 187, 24 UST 2389 (Liability Convention) art. 1 ('For the purposes of this Convention: (a) The term "damage" means loss of life, personal injury or other impairment of health; or loss of or damage to property of States or of persons, natural or juridical, or property of international intergovernmental organizations; (b) The term "launching" includes attempted launching; (c) The term "launching State" means: (i) A State which launches or procures the launching of a space object; (ii) A State from whose territory or facility a space object is launched; (d) The term "space object" includes component parts of a space object as well as its launch vehicle and parts thereof.').

[12]ibid art. 2 ('A launching State shall be absolutely liable to pay compensation for damage caused by its space object on the surface of the earth or to aircraft flight.').

[13]ibid art. 3 ('In the event of damage being caused elsewhere than on the surface of the earth to a space object of one launching State or to persons or property on board such a space object by a space object of another launching State, the latter shall be liable a space object by a space object of another launching State, the latter shall be liable only if the damage is due to its fault or the fault of persons for whom it is responsible.').

[14]ibid art. 4 ('In the event of damage being caused elsewhere than on the surface of the earth to a space object of one launching State or to persons or property on board such a space object by a space object of another launching State, the latter shall be liable a space object by a space object of another launching State, the latter shall be liable only if the damage is due to its fault or the fault of persons for whom it is responsible. 1. In the event of damage being caused elsewhere than on the surface of the earth to a space object of one launching State or to persons or property on board such a space object by a space object of another launching State, and of damage thereby being caused to a third State or to its natural or juridical persons, the first two States shall be jointly and severally liable to the third State, to the extent indicated by the following: (a) If the damage has been caused to the third State on the surface of the earth or to aircraft in flight, their liability to the third State shall be absolute; (b) If the damage has been caused to a space object of the third State or to persons or property on board that space object elsewhere than on the surface of the earth, their liability to the third State shall be based on the fault of either of the first two States or on the fault of persons for whom either is responsible. 2. In all cases of joint and several liability referred to in paragraph 1 of this article, the burden of compensation for the damage shall be apportioned between the first two States in accordance with the extent to which they were at fault; if the extent of the fault of each of these States cannot be established, the burden of compensation shall be apportioned equally between them. Such apportionment shall be without prejudice to the right of the third State to seek the entire compensation due under this Convention from any or all of the launching States which are jointly and severally liable.').

[15]ibid art. 6 ('1. Subject to the provisions of paragraph 2 of this Article, exoneration from absolute liability shall be granted to the extent that a launching State establishes that the damage has resulted either wholly or partially from gross negligence or from an act or omission done with intent to cause damage on the part of a claimant State or of natural or juridical persons it represents. 2. No exoneration whatever shall be granted in cases where the damage has resulted from activities conducted by a launching State which are not in conformity with international law including, in

that state and foreign nationals during their participation in the operation.[16] Article XII provides that damage compensation shall be determined in accordance with principles of international law to the condition if the damage had not occurred.[17]

Finally, Article XXI establishes that if damage from a space object presents a large-scale danger to human life, the States parties and especially the launching state will examine providing assistance to the State that suffered the damage.[18] Article XXI also explicitly states that nothing in the article itself will 'affect the rights or obligations of the States Parties under this Convention'.[19]

4. Registration Convention

The Convention on Registration of Objects Launched into Outer Space ('Registration Convention') provides requirements for the 'State of registry' for space objects.[20] The Registration Convention requires registration of any space object launched into earth orbit or beyond in an appropriate registry, which the Secretary General of the United Nations must be aware of.[21] Article IV provides specifics that each State of registry must furnish to the Secretary General of the United Nations.[22] Such information includes date and location of launch, and the basic orbital parameters.[23]

5. Moon Treaty

The Agreement Concerning the Activities of States on the Moon and Other Celestial Bodies is known at the 'Moon Treaty'.

particular, the Charter of the United Nations and the Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and Other Celestial Bodies.').

[16]ibid art. 7 ('The provisions of this Convention shall not apply to damage caused by a space object of a launching State to: (a) nationals of that launching State; (b) foreign nationals during such time as they are participating in the operation of that space object from the time of its launching or at any stage thereafter until its descent, or during such time as they are in the immediate vicinity of a planned launching or recovery area as the result of an invitation by that launching State.').

[17]ibid art. 12 ('The compensation which the launching State shall be liable to pay for damage under this Convention shall be determined in accordance with international law and the principles of justice and equity, in order to provide such reparation in respect of the damage as will restore the person, natural or juridical, State or international organization on whose behalf the claim is presented to the condition which would have existed if the damage had not occurred.').

[18]ibid art. 21 ('If the damage caused by a space object presents a large-scale danger to human life or seriously interferes with the living conditions of the population or the functioning of vital centres, the States Parties, and in particular the launching State, shall examine the possibility of rendering appropriate and rapid assistance to the State which has suffered the damage, when it so requests. However, nothing in this article shall affect the rights or obligations of the States Parties under this Convention.').

[19]ibid.

[20]Convention on Registration of Objects Launched into Outer Space (opened for signature 14 January 1975, entered into force 15 September 1976) 1023 UNTS 69, 28 UST 695 (Registration Convention) art. 2. ('1. When a space object is launched into earth orbit or beyond, the launching State shall register the space object by means of an entry in an appropriate registry which it shall maintain. Each launching State shall inform the Secretary-General of the United Nations of the establishment of such a registry.').

[21]ibid ('1. When a space object is launched into earth orbit or beyond, the launching State shall register the space object by means of an entry in an appropriate registry which it shall maintain. Each launching State shall inform the Secretary-General of the United Nations of the establishment of such a registry.').

[22]ibid art. 4 ('1. Each State of registry shall furnish to the Secretary-General of the United Nations, as soon as practicable, the following information concerning each space object carried on its registry: (a) name of launching State or States; (b) an appropriate designator of the space object or its registration number; (c) date and territory or location of launch; (d) basic orbital parameters, including: (i) nodal period, (ii) inclination, (iii) apogee, (iv) perigee; (e) general function of the space object. 2. Each State of registry may, from time to time, provide the Secretary- General of the United Nations with additional information concerning a space object carried on its registry. 3. Each State of registry shall notify the Secretary-General of the United Nations, to the greatest extent feasible and as soon as practicable, of space objects concerning which it has previously transmitted information, and which have been but no longer are in earth orbit.').

[23]ibid.

B. Domestic regulatory framework

1. Starlink's FCC approval

The FCC released its Memorandum Opinion, Order and Authorization of SpaceX's Application for Approval for Orbital Deployment and Operating Authority for the SpaceX NGSO Satellite System, and Application for Approval for Orbital Deployment and Operating Authority for the SpaceX NGSO Satellite System Supplement on 28 March 2018.[24] In this order, the FCC authorized SpaceX to launch its satellite system comprising of 4425 satellites to provide fixed-satellite service (FSS) across the globe.[25] Overall the FCC granted in part, dismissed in part, and denied in part the application and supplement.[26] The authorization itself was subject to several conditions which included that it timely provide the Commission with information pursuant to international regulations and the operations in specific frequencies remain in compliance with international regulation.[27] After receiving several comments on both of SpaceX's applications, the FCC ultimately decided to grant SpaceX the license, as it will serve the public interest.[28] The FCC recognized that the grant would 'enable SpaceX to bring high-speed, reliable, and affordable broadband service to consumers in the United States and around the world, including areas underserved or currently underserved by existing networks'.[29] After reviewing the record and public comment, it was clear to the FCC that 'a next-generation space race [was] unfolding,' and that the regulatory frameworks relied upon were dated.[30]

2. FCC framework for small satellites

The FCC responded to the technological innovation of small satellites with a Report and Order *In the Matter of Streamlining Licensing Procedures for Small Satellites* in August 2019.[31] The Communications Act of 1934 requires, 'the issuance of a license for communications to and from the United States or from any U.S. satellite,[footnote omitted]'.[32] According to the FCC, the applications received requesting license or authorization to operate with small satellites represents a growing population of total satellite applications received.[33] The Report and Order in August 2019 responded to the increasing innovation in the small satellite sector by creating a new, optional licensing process for small satellites, as an alternative to 'part 25'.[34] In doing so, the FCC seeks to 'help preserve U.S.

[24]33 FCC Rcd 3391 (4).
[25]ibid para 1.
[26]ibid para. 39 ('Accordingly, IT IS ORDERED, that the Application and Supplemental Application filed by Space Exploration Holdings, LLC (SpaceX) and accepted for filing ARE GRANTED IN PART, DISMISSED IN PART, AND DENIED IN PART, as set forth in this Memorandum Opinion, Order and Authorization, pursuant to section 309(a) of the Communications Act of 1934, as amended, 47 U.S.C. § 309(a).').
[27]ibid para. 40.
[28]ibid para. 4, 5, 7.
[29]ibid para. 1.
[30]33 FCC Rcd 3391 (4), Statement of the Commissioner Jessica Rosenworcel.
[31]34 FCC Rcd. 13077(16), 1–2 ('Recent technological innovation has spurred an increasing use of what have been colloquially termed "small satellites" or "small sats" for a wide variety of missions, ranging from shortterm experimental missions conducting scientific experiments to longer term commercial communications and remote sensing missions.').
[32]34 FCC Rcd. 13077(16), 2 ('The Communications Act of 1934, as amended, requires the issuance of a license for communications to and from the United States or from any U.S. satellite,3 and applications requesting a license or authorization to operate with small satellites represent a growing percentage of the number of satellite applications received by the Commission.').
[33]ibid.
[34]ibid ('Today, we take action to make available a new, optional licensing process for these small satellites. This will enable small satellite applicants to choose a streamlined licensing procedure and thereby take advantage of an

leadership in space-based services and operations'.[35] Additionally, the FCC limits also promotes orbital debris mitigation and the efficient use of spectrum, as well as limits the regulatory burden for applicants and offers potential radiofrequency interference protection.[36]

On 17 April 2018,[37] the FCC released a *Notice of Proposed Rulemaking (NPRM)*, which proposed the modification of the part 25 satellite licensing rules, in order to create a new category of application specific to small satellites.[38] Through the *NPRM*, the FCC sought comment on criteria to define the category, as the proposed streamlined licensing process was based on characteristics typical of small satellite operations.[39] The *NPRM* also sought comment on spectrum-related topics, such as

> typical small satellite frequency use characteristics, how to facilitate compatibility with Federal operations, use of particular spectrum for inter-satellite links by small satellites, and other issues related to operations by small satellites in frequency bands including the 137–138 MHz, 148–150.05 MHz, and 1610.6–1613.8 MHz bands.[40]

Over the comment and reply comment period, the FCC received twenty-three comments and seventeen reply comments.[41]

In the *Report and Order*, the FCC adopted the streamlined version of part 25 for small satellite licensing.[42] The Commission did not propose or adopt any changes to the existing processes in this Order.[43] The adoption of the proposal also was 'to make streamlined

easier application process, a lower application fee, and a shorter timeline for review than currently exists for applicants. We will refer to this alternative as the "part 25 streamlined small satellite process." In so doing, we limit the regulatory burdens borne by applicants and offer potential radiofrequency interference protection for critical communication links, while promoting orbital debris mitigation and efficient use of spectrum. This action will support and encourage the increasing innovation in the small satellite sector and will help preserve U.S. leadership in space-based services and operations.').

[35]ibid.

[36]ibid.

[37]The release occurred less than one month after the FCC issued the Memorandum Opinion, Order and Authorization of SpaceX's *Application for Approval For Orbital Deployment and Operating Authority for the SpaceX NGSO Satellite System* and the *Application for Approval For Orbital Deployment and Operating Authority for the SpaceX NGSO Satellite System Supplement*. 33 FCC Rcd 3391 (4).

[38]34 FCC Rcd. 13077(16), 3 ('On April 17, 2018, the Commission released a Notice of Proposed Rulemaking (NPRM) proposing to modify the Commission's part 25 satellite licensing rules to create a new category of application specific to small satellites.').

[39]ibid ('The Commission sought comment on criteria that would define this new category and proposed that applicants meeting the criteria could take advantage of a simplified application, faster processing, and lower fees, among other things.7 The proposed streamlined licensing process was developed based on the features and characteristics that typically distinguish small satellite operations from other types of satellite operations, such as shorter orbital lifetime and less intensive frequency use.').

[40]ibid ('The NPRM also sought comment on topics related to spectrum use by small satellites. The Commission asked for comment on typical small satellite frequency use characteristics,12 how to facilitate compatibility with Federal operations, 13 use of particular spectrum for inter-satellite links by small satellites,14 and other issues related to operations by small satellites in frequency bands including the 137–138 MHz, 148–150.05 MHz, and 1610.6–1613.8 MHz bands.').

[41]ibid 4 ('Comments on the NPRM were due July 9, 2018, and reply comments were due August 7, 2018. We received 23 comments and 17 reply comments. A list of commenters, reply commenters, and ex parte filings is contained in Appendix C.').

[42]ibid para. 10 ('We adopt here a streamlined version of part 25 for small satellite licensing. Applicants seeking authorization of small satellites can choose to take advantage of this streamlined small satellite process, rather than using the other existing applicable licensing procedures. The goal of this small satellite process is to enable satellites that have shorter missions, less intensive spectrum use, and lower risk of producing orbital debris to be licensed on a streamlined basis.').

[43]ibid para. 13 ('Commenters suggest that the Commission clarify how the proposed rules relate to other existing licensing and authorization processes, particularly those under parts 5 and 97 of the rules. For example, several commenters questioned whether satellite applicants would be prevented from applying for an experimental license under part 5 once the new part 25 rules are adopted.31 We emphasize that all of the existing options for satellite authorization

processing available to entities seeking access to the United States market using a non-U.S.-licensed space station'.[44]

The FCC identified thirteen characteristics of satellite/systems for eligibility for streamlined processing.[45] These thirteen criteria include:

(1) Ten or fewer satellites under a single license. No limitation on the number of applications that may be filed;
(2) Maximum in-orbit lifetime of any individual satellite is six years, including time to deorbit the satellite;
(3) All operations under a license will be completed within six years;
(4) Maximum mass of any individual satellite will be 180 kg, including propellant ('wet mass');
(5) Satellite(s) will be deployed below 600 km altitude or have the capability to perform collision avoidance and de-orbit maneuvers using propulsion;
(6) Satellite(s) will release no planned debris;
(7) Satellite operator has assessed and limited the probability of debris being generated due to an accidental explosion resulting from the conversion of energy sources on board the satellite into energy that fragments the spacecraft;
(8) Probability of in-orbit collision between any satellite and large objects is 0.001 or less as calculated using current NASA software or other higher fidelity model;
(9) Any individual satellite is 10 cm or larger in its smallest dimension;
(10) Satellite(s) will have a unique telemetry marker;
(11) Probability of casualty resulting from uncontrolled atmospheric re-entry of any satellite is zero, as calculated using current NASA software or other higher fidelity model;
(12) Licensees must have the capability to eliminate harmful interference when necessary under the terms of the license or other applicable regulations. In particular, satellites must have the capability for immediate cessation of emissions on telecommand; and
(13) Radiofrequency operations will be compatible with existing operations in the requested frequency bands and not materially constrain future operations of other satellites in those frequency bands.[46]

will remain available, including the existing part 25, part 5 experimental, and part 97 amateur processes.32 No changes to those existing processes were proposed in the NPRM, and none are adopted here.').

[44]ibid para. 14 ('We adopt the NPRM proposal to make streamlined processing available to entities seeking access to the United States market using a non-U.S.-licensed space station, through a petition for declaratory ruling.34 SIA and CSSMA express support for this proposal, provided that the foreign licensed satellite or system is subject to the same requirements as U.S. applicants under the streamlined process and applicable reciprocity market-access requirements under the part 25 process. No commenters disagreed with the proposal. Although we use the term "license" at various points in this Order, the streamlined part 25 process will also be made available to applicants seeking U.S. market access, and conclude that such applicants will be subject to the small satellite streamlined process rules, application and regulatory fees under the new fee categories adopted for small satellites, and the part 25 rules currently applicable to entities requesting to access the United States market using a non-U.S.-licensed space station. We adopt minor revisions to section 25.137 of our rules, addressing non-U.S.-licensed space station application procedures, to add references to the streamlined small satellite process.').

[45]ibid para. 19 ('We summarize below the characteristics of satellites/systems that we have concluded may be eligible for streamlined processing.').

[46]ibid para. 19.

Although several of these criteria overlap with the FCC's separate proceeding, *Mitigation of Orbital Debris in the New Space Age*, the criteria adopted here were developed based on the record developed for the proceeding, 'In the Matter of Streamlining Licensing Procedures for Small Satellites.'[47]

The next section of this paper will discuss the response to the FCC's criterion to limit the applications to ten satellites per application.

a. Number of satellites. The FCC adopted the *NPRM* proposal to limit the number of satellites to be authorized in an individual streamlined part 25 license to ten satellites.[48] The *Report* acknowledges the broad support this number received from commenters, and addresses Boeing's argument to allow up to thirty satellites in an application.[49] In doing so, the FCC states that Boeing's argument to allow up to thirty satellites in a single application would allow a substantially larger constellation than what most small satellite applicants seem to desire.[50] While this would require extensive review, the FCC posits that this could functionally be achieved by applicants simply applying for multiple licenses at time.[51] The FCC, however, declares that the approach does not place a limitation on the number of licenses that can be granted to a single entity, and that it will 'accommodate innovative small satellite system concepts and business models'.[52] The FCC then rejects ORBCOMM and SpaceX's commented concerns applicants 'could manipulate the process and create larger satellite constellations that would not warrant streamlined treatment,' and that 'failure to limit a single company from obtaining licenses for multiple systems runs the risk of greater collision and interference issues, thereby rendering streamlined treatment inappropriate'.[53] The FCC characterizes these concerns as a 'theoretical possibility' which is unlike to occur in practice.[54]

[47]ibid para. 20 ('We note that several of these qualifying characteristics overlap with issues discussed in a separate proceeding addressing the Commission's rules on orbital debris mitigation generally – *Mitigation of Orbital Debris in the New Space Age*. The Commission adopted a Notice of Proposed Rulemaking in that proceeding in November 2018 and comments and reply comments were recently filed. The criteria we adopt here are based upon the record developed specifically in the docket for this proceeding. In the event that we reassess certain orbital debris risks as part of the separate, dedicated orbital debris proceeding, these criteria will be modified as necessary or appropriate to conform to rules that would be generally applicable to Commission-authorized space stations, to ensure regulatory congruity.').

[48]ibid para. 21.

[49]ibid.

[50]ibid. ('And though Boeing argues we should allow up to 30 satellites in a single application,57 that would allow a substantially larger constellation (and require a more intensive review) than what most small satellite applicants appear to desire – and in any event could be functionally achieved by applicants by applying for multiple licenses at the same time.').

[51]ibid.

[52]ibid para. 23 ('We agree and believe the approach we adopt here – which does not place a limitation on the number of licenses that can be granted to a single entity – will accommodate innovative small satellite system concepts and business models.').

[53]ibid para. 24 ('Some commenters such as ORBCOMM and SpaceX express concern that applicants could unfairly manipulate the process and create larger satellite constellations that would otherwise not warrant streamlined treatment. ORBCOMM argues that the Commission should clarify that an applicant cannot file for multiple small satellite system licensees, thereby evading the "more rigorous review of a conventional application." ORBCOMM, SpaceX, and others further argue that failure to limit a single company from obtaining licenses for multiple systems runs the risk of greater collision and interference issues, thereby rendering streamlined treatment inappropriate.').

[54]ibid ('While a theoretical possibility, when viewed in the context of the criteria established for the small satellite licensing process, these concerns are unlikely to be realized in practice. In particular, the six-year orbital lifetime and 600-kilometer maximum altitude (absent propulsion) criteria both correlate with lower collision risk, and the small size of these satellites also correlates with lower risk. Each application will be considered individually and placed on public notice.').

The Analysis section of this paper will discuss the considerations of this specific adopted criteria, as relevant to what it potentially reveals about the FCC interpretation of its and the United States Government's role in small satellite regulation.

C. Fourth Amendment jurisprudence

1. General background: from Warren and Brandeis to Katz

There is no 'right to privacy' provided in the Constitution, including the Bill of Rights. Instead, this concept has developed out of the Fourth Amendment, largely due to the law review article written by Samuel Warren and Louis Brandeis in 1890, 'The Right to Privacy.' Warren and Brandeis proposed that there is in existing of law a principle which can be invoked to protect the privacy of the individual.[55] They assess that this principle is not based in property or contract concepts, and is instead based in the right of an individual to be left alone – the principle of an inviolate personality creates a right to privacy.[56]

The cases of *Olmstead v. United States*, and *Katz v. United States* further developed the doctrinal understanding of the 'inviolate personality'. In *Olmstead*, the Court held that wire tapping did not amount to a search of seizure within the meaning of the Fourth Amendment, because the wire tapping was not a search or seizure of the appellants' person or 'a seizure of his papers or tangible material effects or an actual physical invasion of his house "or curtilage" for the purpose of making a seizure'.[57] The Court did in *Olmstead* did not extend Fourth Amendment to apply the wires because they were not a part of the appellant's house or office.[58]

In 1967, the Supreme Court overruled *Olmstead* with its *Katz* decision.[59] In Katz, the Court the 'trespass' doctrine relied upon in *Olmstead* had since been eroded by subsequent decisions.[60] Instead, the Court relied on the right to privacy as the right to be let alone by other people to mean

> [T]he Fourth Amendment protects individual privacy against certain kinds of intrusion but its protections go further and often have nothing to do with privacy at all … What a person knowingly expses to the public, even in his own home or office, is not a subject of Fourth Amendment protection … But what he seeks to preserve as private, even in an area accessible to the public, may be constitutionally protected.[61]

Justice Harlan's opinion provided the reasonable expectation of privacy test. His understanding was that the Fourth Amendment protection is afforded on a two-fold requirement: first that a person have 'an actual (subjective) expectation of privacy, and second that the expectation be one that society is prepared to recognize as "reasonable"'.[62]

[55]Samuel D. Warren & Louis D. Brandeis, 'The Right to Privacy' (1890) 4 Harv. L. Rev. 193, 197.
[56]ibid 205; 213.
[57]*Olmstead v. U.S.*, 277 U.S. 438, 465 (1928).
[58]ibid.
[59]*Katz v. U.S.*, 389 U.S. 347, 353 (1967).
[60]ibid.
[61]ibid 350–51.
[62]ibid 360–61.

2. Aerial imaging

The late 1980s saw Fourth Amendment questions in regards to aerial imaging. For one scholar, the following three cases led to concerns about satellite surveillance.[63]

In *Dow Chemical Company v. United States*, the Supreme Court held that the Environmental Protection Agency's uses of aerial photography of Dow Chemical Company's plant without a warrant was not a Fourth Amendment search.[64] The Court reasoned that because the Government has 'greater latitude to conduct warrantless inspections of commercial property,' the expectation of privacy in the commercial property differs from that of an individual's home.[65] Further, the Court acknowledged precedent that dictated 'what is observable by the public is observable without a warrant, by the Government as well'.[66] The Court distinguished that surveillance of private property using 'highly sophisticated equipment, not generally available to the public, such as satellite technology, might be constitutionally proscribed absent a warrant,' but the photographs in this case were not revealing of intimate details to raise constitutional concerns.[67] The Court acknowledged that while the aerial photography presented more than what was available by a naked-eye view, it was limited to the outside of the facility's buildings, and thus did not raise constitutional concerns.[68] Accordingly, the photographed area was less like curtilage of a dwelling, and was instead comparable to an open field, as it was of a lawfully accessible nature by persons in aircraft in the public airspace.[69]

In the concurrently decided case, *Californa v. Ciraolo*, the Supreme Court held that the warrantless aerial observation of a fenced-in backyard within the curtilage of a home from an altitude of 1,000 feet was not a violation of the Fourth Amendment.[70] In reviewing law enforcement's use of aerial photography of the marijuana crop growing in an individual's backyard, the Supreme Court applied the Katz two-part test.[71] The Court first distinguished that individual's manifested subjective expectation of privacy was not clearly manifested, and turned to the second part of the inquiry to determine if such an expectation was reasonable.[72] The Court determined that curtilage does not itself bar all police observation, because the observation was available by the naked-eye by the policy from a lawfully operating aircraft.[73] In doing so, the Court acknowledged that 'Fourth Amendment protection of the home has never extended to require law enforcement officers to shield their eyes when passing by a home on public thoroughfares'.[74] According to the Court, such public thoroughfare extended to public airspace, as it served as a public vantage point, lawfully accessed, and which rendered the activities clearly visible.[75] The Court relied on the understanding of Fourth Amendment protection in

[63]Krysten C. Kelly, 'Warrantless Satellite Surveillance: Will Our 4th Amendment Privacy Rights Be Lost In Space?' (1995) 13 J. Marshall J. Computer Info. L. 729, 732–33.
[64]*Dow Chemical Co. v. U.S.*, 106 S.Ct. 1819, 1822 (1986) (United States).
[65]ibid 1826.
[66]ibid.
[67]ibid 1827.
[68]ibid.
[69]ibid.
[70]*California v. Ciraolo*, 106 S.Ct. 1809, 1810; 1813–1814 (1986) (United States).
[71]ibid 1811–12.
[72]ibid.
[73]ibid 1812.
[74]ibid.
[75]ibid 1812–1813.

Katz, that '[w]hat a person knowingly exposes to the public, even in his own home or office, is not a subject of Fourth Amendment protection'.[76]

California v. Ciraolo went on to control the Supreme Court's decision of *Florida v. Riley* in 1989.[77] In this case, Riley took precautions to protect marijuana in his greenhouse against ground-level observation, however, the sides and roof of the greenhouse were left partially open.[78] Under *Ciraolo*, Riley could not reasonably expect for contents of his greenhouse to be immune from the police's viewing from the air at a lawful altitude.[79]

3. Third party doctrine

While Fourth Amendment concerns about satellite surveillance were raised by a scholar in the mid-1990s in light of aerial imaging jurisprudence,[80] these concerns are again relevant today in light of significant changes in technology, widespread presence in space by private companies, data aggregation capabilities, and law. In fact, this issue is ever present in light of *Carpenter v. US*. While the author argued that the logical solution would be to require a warrant for satellite surveillance,[81] this paper explores the third-party doctrines role in such scenario.

The third-party doctrine has developed through Supreme Court cases of *U.S. v. Miller*, and *Smith v. Maryland*, to determine that, 'a person has no legitimate expectation of privacy in information he voluntarily turns over to third parties,' which remains true, 'even if the information is revealed on the assumption that it will be used for only a limited purpose'.[82]

Ciraolo and *Dow Chemicals* found their footing in aerial photography, as it was a method available to the public at the time.[83] In *Kyllo v. U.S.*, the Supreme Court rejected the use of a thermal imaging device used by the Government in order to explore details of a home previously unknowable without physical intrusion, as the technology was not in general public use.[84] Advancements in technology used for surveillance causes questions of Fourth Amendment protection to resurface, based on what expectations remain reasonable. One such question that has begun to resurface Fourth Amendment protections in the context of the third-party doctrine.

In *United States v. Jones*, the Supreme Court held the Government's installation and use of a Global-Positioning-System (GPS) tracking device on an individual's vehicle to monitor the vehicle's movements was a search.[85] The device, '[b]y means of signals from multiple satellites, established the vehicle's location within 50–100 feet, and communicated that location by cellular phone to a Government computer'.[86] Because the GPS device was attached to a vehicle, the Supreme Court recognized it as an 'effect' that was physically occupied under the Fourth Amendment.[87]

[76]ibid 1813.
[77]*Florida v. Riley*, 109 S.Ct. 693, 696 (1989) (United States).
[78]ibid 697.
[79]ibid.
[80]Krysten C. Kelly, 'Warrantless Satellite Surveillance: Will Our 4th Amendment Privacy Rights Be Lost in Space?' (1995) 13 J. Marshall J. Computer Info. L. 729, 732–33.
[81]ibid.
[82]*Carpenter v. US*, 38 S.Ct. 2206, 2216 (2018) (United States).
[83]*Dow Chemical Co. v. U.S.*, 106 S.Ct. 1819, 1824 (1986) (United States).
[84]*Kyllo v. U.S.*, 121 S.Ct. 2038, 2046 (2001) (United States).
[85]*U.S. v. Jones*, 132 S.Ct. 945, 947–49 (2012) (United States).
[86]ibid.
[87]ibid 949.

In 2015, the Eleventh Circuit reviewed whether a court order authorized by the Stored Communications Act (SCA), 18 U.S.C. § 2701 *et seq.* violated an individual's Fourth Amendment rights by compelling the production of a third-party phone telephone company's business records which contained historical cell tower location information.[88] Specifically, under the SCA a U.S. Attorney to obtain a court order requiring 'a provider of electronic communication service … to disclose a record of or other information pertaining to a subscriber to. … such service (not including the contents of communications)'.[89] The Circuit Court applied *Miller* and *Smith* in determining that like the customers in those cases, Davis in this case could assert neither ownership nor possession of the third-party's business records which he sought to suppress, nor would he have a justifiable or reasonable expectation of privacy to the records.[90] Accordingly, the Eleventh Circuit held that the government obtaining the order did not constitute a search and did not violate the Fourth Amendment.[91] Further, the Circuit Court distinguished the physical intrusion at issue in *Jones* to the records and determined it was wholly inapplicable to the case.[92] In doing so, it reviewed Justice Sotomayor's concurrence to the majority opinion in *Jones*, as she raised concern about the GPS monitoring generating 'a precise, comprehensive record of a person's public movements' and gave the government 'unrestrained power to assemble data.'[93] Further, Justice Sotomayor stated that 'it may be necessary to reconsider the premise that an individual has no reasonable expectation of privacy information voluntarily disclosed to third parties,' but did not go further to address it.[94] The Circuit Court then goes on to review Justice Alito's concurrence in *Jones* to draw the distinction of the GPS device on a private vehicle in *Jones*, and the electronic mechanism of the cell tower in *Davis*.[95] *U.S. v. Davis* was appealed to the Supreme Court twice, and certiorari was denied both times.[96] In 2018, the Supreme Court addressed question 'whether the Government conducts a search under the Fourth Amendment when it accesses historical cell phone records that provide a comprehensive chronicle of the user's past movements' in *Carpenter v. U.S.*[97]

a. Carpenter v. U.S.. The 2018 Supreme Court case, *Carpenter v. U.S.*, carved out a narrow exception to the third-party doctrine, but did not overturn it.[98] Two court orders under the Stored Communications Act to obtain cell phone records of Carpenter, disclosing cell-site data.[99] The analysis applied by the Court in *Carpenter v. U.S.* examined two things: first, an individual's reasonable expectation of privacy in information; second, that this information is voluntarily disclosed.[100]

[88] *U.S. v. Davis*, 786 F.3d 498, 500 (11th Cir. 2015) (United States Circuit Court).
[89] ibid 505.
[90] ibid 511.
[91] ibid 513.
[92] ibid 514.
[93] ibid 514 (internal citation omitted).
[94] ibid (internal citation omitted).
[95] ibid (internal citation omitted).
[96] *U.S. v. Davis*, 786 F.3d 498 (11th Cir. 2015) (United States Circuit Court) (*Davis v. U.S.*, 136 S.Ct. 479 (United States) (mem), cert denied); *U.S. v. Davis*, 711 Fed.Appx. 605 (11th Cir. 2017) (United States Circuit Court) (*Davis v. U.S.*, 138 S.Ct. 1548 (United States) (mem), cert denied).
[97] *Carpenter v. US*, 38 S.Ct. 2206, 2211 (2018) (United States).
[98] ibid 2220.
[99] ibid 2212.
[100] ibid 2219–20.

The majority in *Carpenter* held that the third-party doctrine did not apply to the instance of cell-site location information because an individual holds a reasonable expectation of privacy to that information, and it is not voluntarily disclosed.[101] In determining that the government invaded Carpenter's reasonable expectation of privacy in his physical movements, Chief Justice Roberts' majority opinion recalled *Kyllo*. The Court noted that the rule it adopts 'must take account of more sophisticated systems that are already in use or in development'.[102] Further, that while the records in *Carpenter* reflected the state of technology, 'the accuracy of CSLI is rapidly approaching GPS-level precision'.[103]

Chief Justice Roberts then addresses the second factor of the third-party doctrine: voluntary exposure.[104] The majority rations that because cell phones and the services they provide are 'such a pervasive and insistent part of daily life,' having one is 'indispensable to participation in modern society'.[105] Further, the a cell-site records are logged 'by dint of its operation, without any affirmative act on the part of the user beyond powering up'.[106] As a result the location data being generated automatically, a user does not voluntarily 'assume the risk' of turning over physical movements.[107]

Thus, the Supreme Court declined to extend the third-party doctrine *Smith* and *Miller* to the collection of CSLI, and held that Governments acquisition of the records was a search within the Fourth Amendment.[108] In making this narrow decision, Chief Justice Roberts clarified that it does not call into question conventional surveillance techniques and tools, nor does it consider collection techniques involving foreign affairs or national security.[109]

The dissenting opinions adopt the property-based approach to Fourth Amendment doctrine.[110] The dissent of Justice Kennedy, Thomas, and Alito emphasizes that *Miller* and *Smith* were based on the concept that law enforcement had not searched anything belonging to defendants within the meaning of the Fourth Amendment.[111]

Carpenter v. U.S., while demonstrating a narrow exception to the third-party doctrine, raised, through the dissents, concerns about the future of Fourth Amendment jurisprudence and its doctrinal roots in property concepts.

4. Fourth Amendment and national security

While the majority in *Carpenter* did not consider collection methods involving foreign affairs or national security in its analysis or decision, there is some existing jurisprudence at this intersection of privacy and national security.[112]

In 1972, the Supreme Court reviewed the issue of the President's power to authorize electronic surveillance without prior judicial approval in U*.S. v. United States District*

[101]ibid.
[102]ibid 2219 (internal citations omitted).
[103]ibid.
[104]ibid 2220.
[105]ibid 2220 (internal citations omitted).
[106]ibid.
[107]ibid (internal citations omitted).
[108]ibid.
[109]ibid 2220.
[110]ibid 2224 (J. Kennedy, J. Thomas, J. Alito, dissenting).
[111]ibid 2227 (J. Kennedy, J. Thomas, J. Alito, dissenting).
[112]ibid 2220.

Court ('*Keith*').[113] The Court had to balance the 'Government's right to protect itself from unlawful subversion and attack' against the citizen's right to privacy against unreasonable Government intrusion.[114] In doing so, the Supreme Court held that there is no domestic surveillance exception to the warrant requirement of the Fourth Amendment.[115] Similarly narrow in its holding, the Court clarified that it did not address and expressed no opinion on the activities of foreign powers or their agents.[116] Further, it recognized that domestic security surveillance differs from 'ordinary' crime, and concluded that 'the focus of domestic surveillance may be less precise than that directed against more conventional types of crime'.[117]

In 2002, the United States Foreign Intelligence Surveillance Court of Review, determined that the Foreign Intelligence Surveillance Act (FISA), as amended in 2001 by the Patriot Act, did not require 'the government to demonstrate to the FISA court that its primary purpose in conducting electronic surveillance is not criminal prosecution [emphasis omitted]'.[118] The Patriot Act amendment disapproved the primary purpose test, resulting in FISA applications to be granted even if 'foreign intelligence' is only a significant and not primary purpose; another purpose could be the primary purpose.[119] Accordingly, the amendment eliminated the need for the FISA court to balance 'the relative weight the government places on criminal prosecution as compared to other counterintelligence responses'.[120] Further, the government meets the statutory test if the purpose is broader than criminal prosecution, including stopping an ongoing conspiracy.[121] In considering whether the amended statute was consistent with the Fourth Amendment, the FISA Court of Review applied the balancing test of *Keith* to eventually determine that the amended FISA statute is constitutional 'because the surveillances it authorizes are reasonable'.[122]

In 2008, The FISA Court of Review reviewed an expired amendment to FISA, known as the Protect America Act of 2007 ('PAA').[123] PAA authorized the United States 'to direct communications service providers to assist it in acquiring foreign intelligence when those acquisitions targeted third persons (such as the service provider's customers) reasonably believed to be located outside the United States'.[124] The FISA Court of Review 'weighed the nation's security interests against the Fourth Amendment privacy interests of United States persons' to determine that the directives were lawful and compliance with them was obligatory.[125] The FISA Court of Review recalled that the Supreme Court had reserved the question of a foreign intelligence exception in *Keith*, and purports that the Court has recognized a comparable exception with 'special needs cases'.[126]

[113] *United States v. United States District Court for the Eastern District of Michigan, Southern Division (Keith)*, 407 U.S. 297, 299 (1972).
[114] ibid.
[115] ibid 321.
[116] ibid.
[117] ibid 322.
[118] *In Re Sealed Case*, 310 F.3d 717, 736 (F.I.S.C. 2002) (United States Foreign Intelligence Surveillance Court of Review).
[119] ibid 734.
[120] ibid 735.
[121] ibid 735.
[122] ibid 746.
[123] *In re Directives Pursuant to Section 105B of the Foreign Intelligence Surveillance Act*, 551 F. 3d 1004, 1006 (F.I.S.C. 2008) (United States Foreign Intelligence Surveillance Court of Review).
[124] ibid.
[125] ibid.
[126] ibid 1010.

A special needs exception excuses the Warrant Clause when the purpose of the government – law enforcement – action went 'beyond the normal need for law enforcement,' making 'the warrant and probable-cause requirement[s] impracticable'.[127] The FISA Court of Review then considered 'whether the reasoning of the special needs cases applies by analogy' and concluded that it does justify a foreign intelligence exception to the warrant requirement.[128] This exception was for surveillance out of a national security purpose and 'directed at a foreign power or agent of a foreign power reasonably believed to be located outside the United States'.[129] The FISA Court of Review then determined that the PAA, as applied, constituted a 'sufficiently reasonable exercise of government power to satisfy the Fourth Amendment'.[130] In its Fourth Amendment analysis, the Court determined the reasonableness of the particular government action by considering the totality of circumstances.[131]

5. Starlink privacy policy

Starlink's privacy policy website page provides an introduction which outlines that Statlink does not sell personal information, but it is required to disclose what is collected and how it is used.[132] Further it provides an exception that if the Starlink services are through an organization's subscription, the use should ask the organization's administrator for information about how personal information is processed.[133] Starlink collects seven different types of personal information, including: identity data, contact data, profile data, financial data, transaction data, website technical data, and consumer technical data.[134] In the second section 'How and Why Do We Use Personal Information,' one of the purposes of use given for the seven types of personal data collected includes, 'To comply in good faith with applicable laws, legal processes, and lawful government requests.'[135] The third section, 'With Whom Do We Share Personal Information' details that Starlink does not sell personal information, but may share it with three categories of people: Starlink's service providers; government agencies, regulators, and professional advisors; and organizations involved in business transfers.[136] The second of these categories, 'Government Agencies, Regulators and Professional Advisors' details that '[w]here required by applicable law, we may also need to transfer your personal information to government agencies and regulators (e.g. tax authorities, courts, and government authorities)'.[137] This section concludes by providing:

> As a Starlink services customer, you may share information with third parties (for example, when you send an email or communicate with a third-party website). In this context, we

[127]ibid 1010 (internal citations omitted).
[128]ibid 1011.
[129]ibid.
[130]ibid 1012.
[131]ibid 1012 (internal citations omitted).
[132]'Starlink Privacy Policy' (Starlink, last updated July 1, 2021) <https://www.starlink.com/privacy-policy> accessed January 2021.
[133]ibid.
[134]ibid.
[135]ibid.
[136]ibid.
[137]ibid.

are not sharing personal information; you are using our service to share data and we are merely connecting you to the Internet.[138]

Accordingly, the Starlink Privacy Policy seeks to be provide its consumers with valuable information about their personal data and its uses.

II. Analysis

A. United States liability pursuant to international and domestic regulation

1. International law and FCC regulation

Regardless of the intent or strategy behind such criteria implemented by the FCC, international law creates liability for the United States government through the Liability Convention.[139] Under Article III of the Liability Convention, the 'launching state' is liable for damage caused elsewhere than the surface of the earth, and will only be extended to the launching state if the damage is 'due to its fault or the fault of persons for whom it is responsible'.[140]

In an April 2018 FCC Enforcement Advisory, the FCC acknowledged that it must comply with licensing obligations per the International Telecommunication Union (ITU) Radio Regulations (ITURR).[141] As treaty provisions are binding on the United States, the ITURR requires that 'no transmitting station may be established or operated by a private person or by any expertise without a license or on behalf of the government of the country to which the station in question is subject'.[142] The FCC then acknowledged The Communications Act of 1934, as amended, which requires 'the issuance of a license for communications to and from the United States or from any U.S. satellite, and provides the FCC with authority to take actions to implement the [ITURR]'.[143] Finally, the FCC recognized that its licensing is an important aspect of ensuring that the U.S. satisfies its treaty obligation, and concludes with that statement '[o]perations without a license are subject to the full range of FCC enforcement actions'.[144] While there is a specific range of FCC enforcement actions here, it is noteworthy that the FCC began its Enforcement Advisory, entitled 'COMPLIANCE WITH SATELLITE COMMUNICATIONS LICENSING REQUIREMENTS IS MANDATORY AND FAILURE TO COMPLY CAN RESULT IN ENFORCEMENT ACTION,' with the governing international and domestic law, with its purpose to ensure compliance.[145] Further, the FCC begins this Enforcement Advisory with the reminder that '[t]he Commission has previously issued guidance on obtaining licenses for small satellites,' and cites to a Public Notice from 2013, establishing that

[138]ibid.
[139]Convention on International Liability for Damage Caused by Space Objects (opened for signature 29 March 1972, entered into force signed September 1972) 961 UNTS 187, 24 UST 2389 (Liability Convention) art. 1; 3.
[140]ibid.
[141]33 FCC Rcd 3592 (5), 1 ('The International Telecommunication Union (ITU) Radio Regulations are treaty provisions binding on the United States, and require that no transmitting station may be established or operated by a private person or by any enterprise without a license by or on behalf of the government of the country to which the station in question is subject').
[142]ibid 1–2; International Telecommunication Union (ITU) Radio Regulations, No. 18.1 (2015).
[143]ibid 2; 47 USC § 301 (d), (f); 47 USC § 303(r).
[144]ibid 2 (citing the 'Outer Space Treaty' art. 6, adopted Oct. 10, 1967, 18 U.S.T. 2410, 610 U.N.T.S. 205).
[145]ibid 1.

entities have been on notice of this guidance for the five years prior to the issuance of this advisory.[146] The FCC then addresses satellite authorization by a foreign administration:

> Satellites authorized by an administration other than the United States do not require any FCC approval if earth station operations are exclusively outside the United States. If earth station operations in the United States are contemplated, the applicant for a U.S. earth station should be prepared to identify the administration authorizing the space station operations under Article 18.1 of the ITU Radio Regulations, and the country that has registered or intends to register the satellite pursuant to the U.N. Outer Space Treaty.[147]

The first proposition of the above guidance clarifies that the FCC's authorization for satellite launches us not required where the earth stations are 'exclusively outside of the United States'.[148] Conversely, this proposition requires FCC authorization for satellite launches taking place in the United States.

As applied to SpaceX's Starlink program, FCC authorization is required. According to RocketLaunch.Live, the compiled Starlink Launch Schedule provides that the Startlink satellite launches from 23 May 2019 (the first Starlink launch) to 24 November 2020 (the sixteenth launch) all occurred within the United States.[149] The launches vary between Cape Canaveral AFS, Florida, and Kennedy Space Center, Florida; all occurring, however, in the United States.[150]

Article I of the Liability Convention presents the following definition for 'launching state:'

> (c) The term 'launching State' means: (i) A State which launches or procures the launching of a space object; (ii) A State from whose territory or facility a space object is launched; [i]n the event of damage being caused elsewhere than on the surface of the earth to a space object of one launching State or to persons or property on board such a space object by a space object of another launching State, the latter shall be liable a space object by a space object of another launching State, the latter shall be liable only if the damage is due to its fault or the fault of persons for whom it is responsible.[151]

Under Article (1)(c)(ii) of the Liability Convention then, the United States is the launching state. Applying Article III of the Liability Convention, the United States government-through authorization of the FCC for the launches taking place within the United States-is liable for 'damage being caused elsewhere than the surface of the earth' to the objects specified.[152]

The FCC's Enforcement Advisory also highlights that U.S. satellite operators should be aware that for operations launching via earth stations outside the United States the regulatory authorities in other countries may first require a U.S. space station authorization

[146]ibid 1. ('The Commission has previously issued guidance on obtaining licenses for small satellites' (citing Guidance on Obtaining Licenses for Small Satellites, Public Notice, DA 13-445, 28 FCC Rcd 2555 (Mar. 15, 2013), available at https://www.fcc.gov/document/guidance-obtaining-licenses-small-satellites.)).

[147]ibid 3.

[148]ibid.

[149]'Launch Schedule for Series: SpaceX Starlink and Including all past launches' (RocketLaunch.Live) <https://www.rocketlaunch.live/> (search 'Select Filter' for Provider/Vehicle Designation of 'SpaceX/Falcon 9'; click 'Series: SpaceX Starlink'; click 'Show Past Launches').

[150]ibid.

[151]Convention on International Liability for Damage Caused by Space Objects (opened for signature 29 March 1972, entered into force signed September 1972) 961 UNTS 187, 24 UST 2389 (Liability Convention) art. 1.

[152]Convention on International Liability for Damage Caused by Space Objects (opened for signature 29 March 1972, entered into force signed September 1972) 961 UNTS 187, 24 UST 2389 (Liability Convention) art. 3.

before issuing an earth station authorization.[153] Although there is more than one state involved here, the Liability Convention would apply, binding the launching state for liability.

The Enforcement Advisory also provides that 'a U.S. satellite may require more than on U.S. authorization, depending on the proposed operations and launch scenario'.[154] While additional authorizations may be required, they do not guarantee approval of an FCC license.[155]

Finally, the notice addresses where FCC satellite authorizations may be found-specifically where authorizations under Part 25 and Part 5 can be located.[156] The August 2019 *Report and Order* introduced an alternative small satellite approval process for part 25.[157] While it creates an alternative, it does not establish this process to be outside of its satellite authorization process. Thus, the aforementioned analysis of the Liability would apply similarly to the streamlined version of part 25 for small satellite licensing.

2. Orbital debris mitigation

ORBCOMM and SpaceX raised concerns about the increased risk of collision by saturating LEO with small satellites, should a single company obtain licenses for multiple systems.[158] The FCC's response to this concern was to characterize it as a 'theoretical possibility' and unlikely to occur, 'when viewed in the context of the criteria established for the small satellite licensing process'.[159] This might not be a justified response based on the FCC's specific actions.

The FCC particularly relies on its criteria for a six-year orbital lifetime, and 600-kilometer altitude limitation, as well as the satellites small size as it correlates with lower risk.[160] The FCC's corresponding footnote reads:

> As discussed infra, section III.B.6, the issue of aggregate collision risk for constellations is the subject of a separate proceeding that may result in specification of additional criteria for evaluating all satellite systems, including those qualifying for the small satellite licensing process.[161]

This footnote, however, does not specifically provide a correlation between the criteria and risk for collision.[162] While the *Report* itself mentions the *Mitigation of Orbital Debris in the New Space Age Initiative* process, the FCC seemingly downplays the risk of collision in addressing ORBCOMM and SpaceX's concerns here.

Alternatively, the FCC places a great deal of confidence in these two criteria, considering the increased risk of international liability the U.S. government binds itself to with every authorization of small satellite deployment. It is possible that the FCC does not believe it is bound to this international liability by way of the criteria for approval it is establishing here. Instead of being subject to liability, the FCC may believe it is creating

[153] 33 FCC Rcd 3592 (5), 2.
[154] ibid.
[155] ibid.
[156] ibid 3.
[157] 34 FCC Rcd. 13077(16), para. 9.
[158] ibid para. 24.
[159] ibid.
[160] ibid.
[161] ibid FN 65.
[162] *See* ibid.

a buffer to such liability by implementing the specific criteria identified and adopted in this *Report and Order*, as if they are contractual terms. The contradiction to that approach, however, is the fact that the FCC's rule to limit applications to ten satellites at a time must be implemented to address some concern held by the Commission around the volume of small satellites entering LEO, especially as the application volume increases. The FCC in adopting this limiting criteria is choosing to have at least some control over developing mega-constellations.

3. FCC regulating the small satellite market and assessing opportunity cost

At some level, the FCC is concerned about authorizing substantially larger constellations at once, and seeks to create a check on this possibility by creating the adopted limitation of ten small satellites per application.[163] The explanation offered for this in the next paragraph is for the FCC to review each application to ensure that operations continue to meet the qualifications, and not materially constraining the other operations in the requested frequency band.[164] At some level, the FCC engages in a market regulating function through conducting an opportunity cost assessment of the applications presented.[165] Further, through licensure, the FCC is allocating the United States government's property rights by approving private entities to occupy spectrum in low-Earth orbit (LEO).[166] While the FCC perceives that the approach is not placing a limitation on the number of licenses that can be granted to a single entity, it embraces the fact that the NPRM itself was a response to a market disruption.[167]

At some point, the approval process becomes a matter of opportunity. While the Commission had its reservations about Starlink, it ultimately approved the innovation because it would enable 'high-speed, reliable, and affordable broadband service' be brought to consumers not only in the United States, but also around the world, including areas traditionally underserved by these services.[168] The FCC had a choice: it could either approve, with reservation, SpaceX's application and bring broadband internet services to every corner of the globe, or it could deny or limit the project in ways that would limit its effectiveness. With the onset of Starlink as a market disruption, the FCC chose to change the status quo, globally.

4. One cohesive regulatory framework

The international and domestic regulatory frameworks operate in synergy. The FCC is bound by international law to be a liability for launches that take place through its authority in the United States. Through the NPRM, the FCC is beginning to implement additional regulation to adequately account for the growing volume of material in LEO.

[163] 34 FCC Rcd. 13077(16), para. 21.
[164] ibid para. 22.
[165] *See generally* Henry N. Butler, et al., *Economic Analysis for Lawyers* (3d ed. 2014) 635. (Opportunity cost is defined as '[t]he highest valued alternative that must be sacrificed as a result of choosing among alternatives. The value placed on opportunities foregone in choosing to produce or consume scarce good.').
[166] *See generally* Henry N. Butler, et al., *Economic Analysis for Lawyers* (3d ed. 2014) 18–19. ('A property right is a socially enforced right to determine and control the use of an economic good. … In general, the granting of property rights confers two types of economic benefits, static and dynamic. … The transferability of property rights facilitates the maximization of the static and dynamic benefits of well-defined property rights.').
[167] 34 FCC Rcd. 13077(16), para. 23; para 1.
[168] 33 FCC Rcd 3391 (4) para. 1.

B. Third party doctrine application analysis

1. Carpenter majority opinion

While the majority opinion of *Carpenter v. US* carved out a narrow exception in light of developing technologies, it is unclear what the third-party doctrine's application in light of the new space race will look like. Eventually, the Supreme Court will have to make a clear decision on the third-party doctrine's application.

2. Carpenter property concepts

The property concepts raised by the dissents in *Carpenter v. U.S.* can be contrasted with the question of enterprise rights raised by one author in the 2018 Missouri Law Review, 'Into the Final Frontier: The Expanse of Space Commercialization.' In discussing harvesting rights in space, this paper defines 'enterprise rights' as 'entitlements of private entities to operate in an unowned space and to collect and use the resources gained'.[169] An argument proposed by the author suggests that although the Outer Space Treaty's non-appropriation principle bars exclusive occupation of a location, enterprise rights would make it so commercialization of space is fair 'by preventing monopolies of desirable areas and resources'.[170] Then the author explains that such arrangements are already in use for satellites, as States must register orbital positions in geostationary orbit in order to launch and operate the satellites.[171] Thus, the author concludes that States 'are currently exercising enterprise rights through their use of particular spots in space'.[172]

Pursuant to the 'Opportunity Costs' discussion, are States utilizing their property rights or enterprise rights by authorizing small satellites to occupy space in LEO? And how does the Liability Convention interact with these principles?

By its existence, the Liability Convention implies that private entities must be subject to State sovereignty when engaging in space activity. By application of the Liability Convention, it the United States government by way of domestic and international regulation would be responsible for specific activity of private satellites in LEO. Through the domestic and international licensure process, the United States government is utilizing its property right by authorizing small satellites. Whether the United States government is applying its property rights or enterprise rights, the result is the same: it is the one authorizing the private entity to occupy and exist in orbit.

Applying the property right, the United States government is granting by license to the private entity the right to exist at a specific location. This is as far as the property right should extend. While there might be an argument that perhaps that sovereign state owns access to resources that specific location is privy to, the international registration requirement forfeits that right; instead, the State is bound to international sovereignty.

Applying the enterprise right, the private commercial entity operating in a specific location would own the resources collected and gained in that location. In the context of Starlink, this means that each of the satellites in orbit owns the data and information it is collecting through operation of its services as an internet service provider (ISP). In this regard, Starlink would own information traditionally collected by ISPs through their

[169]Rachel Mitchell, 'Into the Final Frontier: The Expanse of Space Commercialization' (2018) 83 Mo. L. Rev. 429. 444.
[170]ibid.
[171]ibid.
[172]ibid.

routine operation. Because the United States government is authorizing through licensure the operation of the satellites in, it in itself is not operating at the specific location in orbit. The utilization of its property right effectively removes it from engaging in operation or as an enterprise in that location. The result, however, is the same. By granting licensure for Starlink or satellites to operate, the United States is granting its right to the corporation. Since the organization is operating under a license, whatever the organization is able to generate from the location as part of its general operation, belongs to the organization. Thus, under the current regulatory framework, the United States government could not simply access Starlink's business records. Instead, as indicated in Starlink's Privacy Policy, it would have to make a lawful request for consumer records.

3. Does the third-party doctrine exception from Carpenter *apply?*

Similar to the banks in *Miller*,[173] Starlink is a third-party. In the case of Starlink, it provides in its Privacy Policy the personal data that it is collecting, but not the full extent of the data collected by each small satellite. Under the majority opinion's *Carpenter* analysis, it must first be determined if consumers of the internet service have a reasonable expectation of privacy recognized by society, and if the information is being voluntarily disclosed. Pursuant to the *Carpenter* precedent, the existence of a reasonable expectation of privacy will depend on the specific data in question – including if it is revealing of intimacies of life. This is further complicated, however, by the aerial imaging cases of *Dow Chemicals, Ciraolo* and *Riley*. While this could result in any accessible from a lawful air space to be available to law enforcement, the check, however, on the naked eye, and open fields doctrines is *Kyllo*. Since satellite imaging technology are not available for general public use, [174] it would likely be limited by *Kyllo* if the imaging capabilities are beyond that of GoogleMaps.

The voluntary disclosure argument, however, is analogous to *Carpenter*. The Court recognized the cell-site location information was being collected continuously, and a cell phone is a vital part of daily life.[175] Similar to cell-site location information, data related to internet usage should be treated the same way as it is collected and logged 'by dint of its operation, without any affirmative act on the part of the user beyond powering up'. [176]

The holding in *Carpenter* was explicit to not disrupt the current operation of the third-party doctrine, as it carved a narrow exception. Accordingly, the third-party doctrine would then apply as it did in *Smith* and *Miller*. Any consumer usage records would be similar to business records collected by a bank or telephone company. Starlink's Privacy Policy acknowledges this, as it maintains that it may have to disclose information in response to 'lawful government requests'. The analysis in *Carpenter*, however, may have moved from the black and white application of the third-party doctrine to one with much more grayspace and uncertainty.

4. How does it apply under the national security privacy framework?

Carpenter simply carved a narrow exception but did not seek to disrupt the current foreign affairs or national security applications of the law. Accordingly, under *Keith*, *In*

[173]*Carpenter v. US*, 38 S.Ct. 2206, 2227 (2018) (United States).
[174]*Kyllo v. U.S.*, 121 S.Ct. 2038, 2046 (2001) (United States).
[175]*Carpenter v. US*, 38 S.Ct. 2206, 2216–20 (2018) (United States).
[176]ibid 2220.

Re Sealed, and *In Re Directives*, the government could operate under a foreign surveillance exception in seeking data from Starlink, and Starlink would be required to comply. While *In Re Directives* dealt with a Communications company, Starlink would similarly be bound by federal regulation, such as the Stored Communications Act to act accordingly.

This could especially be true if the United States is to encounter similar threats as those once posed by the events of September 11, 2001. The PAA was passed in its aftermath, and although it expired, the case law remains. The framework for the government to similarly conduct such warrantless surveillance has been set and validated by *In Re Sealed* and *In Re Directives* by the FISA Court.

One alternative would be to do away with such precedent and that of the third-party doctrine, and rely on the reasonable expectation of privacy test by *Katz*. In the face of foreign threat, however, this would still be unlikely.

Conclusion

Space technologies and operations are rapidly developing. In order to keep up with potential violations of an individual's reasonable expectation of privacy, Fourth Amendment jurisprudence will have to evolve. As other questions of surveillance and third-party doctrine are addressed through advancements in technology, the courts will have to address questions of individual rights in the final frontier.

Disclosure statement

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